Address Verification Manual

BASE24-pos®

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Preface

The ***BASE24-pos Address Verification Manual*** is designed to give readers a thorough understanding of address verification processing support provided in both the standard BASE24-pos product and the BASE24-pos add-on Address Verification module.

Audience

This manual targets three types of readers. It is intended for the individual seeking an overview of BASE24-pos Address Verification processing. It is also intended for BASE24 operators responsible for configuring and maintaining BASE24-pos Address Verification. Finally, this manual is intended for host site personnel interested in identifying the impacts of BASE24-pos Address Verification processing on their system.

Prerequisites

This manual is a stand-alone document providing a complete overview of BASE24-pos Address Verification and all the information necessary to establish and maintain address verification processing in a BASE24-pos system. The address verification processing procedures are presented in this manual for a reader who is already familiar with standard BASE24-pos and host procedures.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers that do not need the additional detail and at the same time provide the source for readers that require the additional information. This manual contains references to the following BASE24 publications:

- The ***BASE24 Tokens Manual*** contains descriptions of all tokens that can be carried in the internal message and procedures for adding and updating records in the Token File (TKN).
- The ***BASE24-pos Transaction Processing Manual*** contains a description of the BASE24-pos Standard Internal Message (PSTM) portion of the internal message.
- The ***BASE24-pos Settlement and Reporting Manual*** contains cardholder activity and retailer activity report samples.
- The ***BASE24-pos Files Maintenance Manual*** contains information on how to peruse the POS Transaction Log File (PTLF).
- The ***BASE24 External Message Manual*** contains a description of all elements in the BASE24 ISO External Message.
- The ***BASE24 Base Files Maintenance Manual*** contains detailed information on the Cardholder Authorization File (CAF), Card Prefix File (CPF), External Message File (EMF), Host Configuration File (HCF), and the Institution Definition File (IDF).
- The ***BASE24 Files and Record Structures Reference Manual*** provides general information about the structure of BASE24 files and records and instructions for using the DDLDOC program to obtain detailed record layouts.

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The ***BASE24-pos Address Verification Manual*** is divided into the following sections.

Section 1, "Introduction," provides an overview of address verification and how BASE24-pos supports it.

Section 2, "BASE24-pos Implementation," provides information for implementing BASE24-pos Address Verification with or without the add-on module that allows the BASE24-pos Authorization module to perform address verification.

Section 3, "Host Impacts," identifies the impacts that BASE24-pos Address Verification has on host processing.

"Conventions Used in This Manual" follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, PS-AE110-01 for the ***BASE24-pos Address Verification Manual***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Dec-2009 for December, 2009) indicates the issue of the manual. The software release information (for example, R6.0v10 for release 6.0, version 10) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

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Conventions Used in This Manual

This section explains how field descriptions are documented in this manual.

Field Descriptions

Each field appearing on a file maintenance screen is listed by name and then described. Field descriptions briefly summarize the contents, purpose, and permissible values, as shown in the following example taken from the Cardholder Authorization File (CAF).

Example

ADDRESS — The cardholder billing address used when performing address verification. All information that is numeric in nature must be entered as numeric data (e.g., First Street must be entered as 1st Street).

Field Length:	1–20 alphanumeric characters
Required Field:	No
Default Value:	No default value
Data Name:	CAF.AVCAF.ADDR

Explanation

Each field description is completed by one or more of the following items of information:

Item	Description
Example	Illustrates a possible entry for the field to further clarify the value or values that can be entered.

Item	Description
Field Length	Specifies the size of the field and the type of characters that can be entered. This length refers to the field and valid values for the file maintenance screen, not the field in the DDLs. Possible values are alphabetic, alphanumeric, hexadecimal, and numeric. The term *alphanumeric* includes all alphabetic, numeric, and special characters that can be entered from a keyboard without using a control sequence. The term *hexadecimal* includes all numbers and the letters A through F. When a field value cannot be modified by the operator, the field length is System-protected.
Occurs	Indicates the number of times the field can be displayed on the screen. This information is provided only when the field can be displayed multiple times.
Required Field	Specifies whether a value has to be entered in the field. Possible values are Yes and No. Some fields are required only under certain conditions. In this case, the entry is Yes, followed by the conditions that determine when the field is required.
Default Value	Specifies the value that is automatically placed in the field when the screen is first displayed or when the **F8** key is pressed to clear the screen.
Data Name	Provides the DDL name associated with the field appearing on the screen. Data names are included in the documentation to assist in communicating screen and field issues to your technical staff. Note that screen data is not always stored in the BASE24 database as it appears on the screen or as it is described in the field description. If you need information on how screen data is actually stored, consult the DDLs.

1: Introduction

This section provides an overview of address verification and how the BASE24-pos product supports it. The section identifies the processing options the BASE24-pos product offers and the issues to consider when implementing address verification.

Address Verification

Address verification is intended to assist merchants in controlling losses that can result from credit card fraud during transactions where the cardholder cannot be readily identified. Address verification gives these merchants additional information to use in confirming that a customer is entitled to use an account when it is not possible to physically examine the card and the customer's signature.

BASE24-pos Support

The ***Address Verification Service Interface Specification*** from Visa U.S.A., Inc. Switching and Authorization Services and the ***Operations Bulletin*** from MasterCard International, Inc. define the basic level of address verification support a processor must provide. A processor must be able to accept a transaction from the VisaNet Interchange or the Banknet Interchange, verify the address and ZIP code information it contains, and return an address verification status code in the response. The processor can perform the address verification or pass the information to a host or interchange that performs the verification.

The standard BASE24-pos product accepts address verification transactions from the VisaNet Interchange and the Banknet Interchange and uses the BASE24-pos Host Interface process or BASE24 Interchange (BIC) Interface process to pass the information to a host or another interchange for verification. When the add-on BASE24-pos Address Verification module is purchased and installed, the BASE24-pos Device Handler/Router/Authorization process also verifies addresses using information maintained in an additional segment of the Cardholder Authorization File (CAF).

The BASE24-pos product also provides address verification support in the Merchant Host Interface (MHI). However, address verification support is not currently provided for devices attached to BASE24-pos VISA I, VISA II, Standard (SPDH), Hypercom (HPDH), NCR/NDP, CRT Administration, or CRT Authorization Device Handlers.

When the add-on BASE24-pos module is used to perform address verification, the address information is contained in an additional segment of the CAF. Since this information is in the CAF, the BASE24-pos Device Handler/Router/Authorization process can only perform address verification when the Positive Authorization (CAF), Positive Balance Authorization (CAF/PBF), or Parametric Authorization (CAF/PBF/CAPF) method is used.

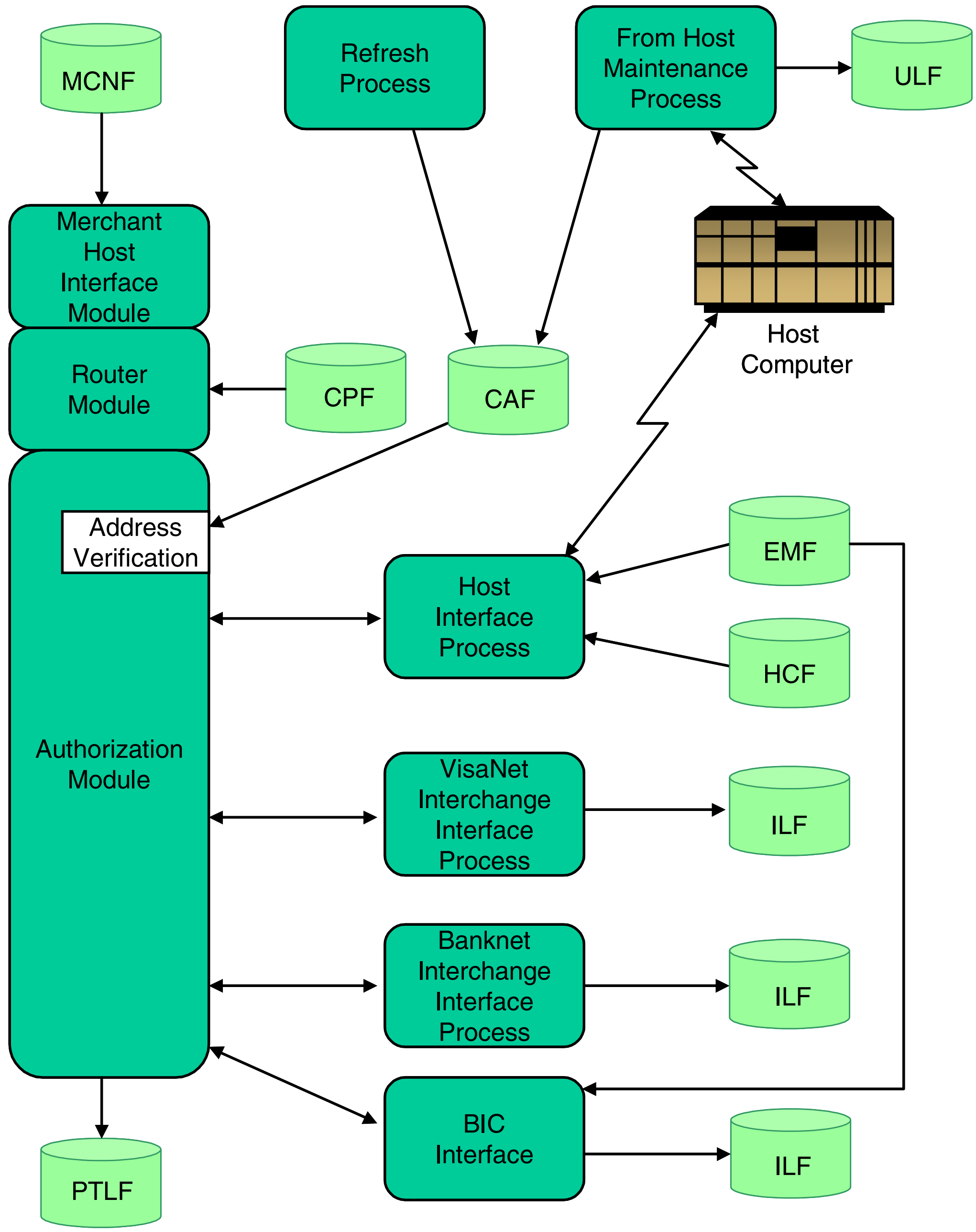
Address verification processing is optional with all BASE24-pos transactions, including card verification requests, based on the presence or absence of address information in the message. The BASE24-pos Device Handler/Router/ Authorization process verifies the address information contained in an incoming message or passes the information to a host or another interchange. Address verification processing is omitted if an incoming message does not contain address information. The BASE24-pos Device Handler Router/Authorization process does not modify the result of address verification performed by a host or another interchange.

Note: References to transaction authorization in this manual include card verification requests, as well as financial transactions.

The following BASE24 interfaces include support for address verification:

- BASE24-pos Merchant Host Interface (MHI)
- VisaNet ISO Interface
- Banknet ISO Interface
- ANSI and ISO BASE24-pos Host Interface
- ANSI and ISO BASE24 Interchange (BIC) Interface

The diagram on the following page illustrates the processes and files involved in standard and add-on BASE24-pos Address Verification transaction processing. Super Extract, Reporting, and BASE24 requesters and servers supporting BASE24 CRT access are also involved, but do not appear on the diagram. The remainder of this section describes each component.


MCNF
Refresh Process
From Host Maintenance Process
ULF
Merchant Host Interface Module
Host Computer
Router Module
CPF
CAF
Address Verification
EMF
Host Interface Process
HCF
Authorization Module
VisaNet Interchange Interface Process
ILF
Banknet Interchange Interface Process
ILF
BIC Interface
ILF
PTLF

Merchant Responsibilities

BASE24-pos Address Verification provides a status code indicating how the address information entered by the merchant with a transaction compares to address information carried in a database. The database can be maintained on BASE24-pos, a host, or an interchange. The status code does **not** change the authorization result.

Because a transaction is not automatically declined due to a discrepancy in address information, the merchant must have access to the address verification status code in order to take advantage of BASE24-pos Address Verification support. If a transaction is approved but the address verification status code indicates a questionable or invalid address, the merchant must determine whether the transaction is acceptable or should be reversed. If a reversal is necessary, the merchant must manually initiate the reversal transaction.

AVS Condensing Algorithm

The BASE24-pos Device Handler/Router/Authorization process, the VisaNet Interchange, and the Banknet Interchange use the Address Verification Service (AVS) Condensing algorithm to compare addresses submitted with transactions against addresses in an address verification database. The resulting status codes are then used by the merchant as one step in the process of determining the customer's identity. Users should verify the algorithm and status codes used by hosts or other interchanges performing address verification for consistency.

The processor performs the following tasks for address verification processing using the VisaNet algorithm:

1. Computes an interim address verification value from the ZIP code and up to the first five numeric characters before the first alphabetic character or space of the address carried in its address verification database.
2. Computes an interim address verification value from the ZIP code and up to the first five numeric characters before the first alphabetic character or space carried in the incoming message. The backslash (\), forward slash (/), number sign (#), and hyphen (-) characters within the first five numeric characters are ignored.
3. Compares the two interim address verification values and determines the appropriate status code value.

Interim Address Verification Values

The interim address verification values consist of two components—the ZIP code and the first five digits from the address.

The ZIP code for a U.S. cardholder must be five or nine digits in length. The ZIP code for a non-U.S. cardholder can be one to nine alphanumeric characters in length and can include spaces. If the ZIP codes being compared are of different lengths, the comparison includes only the first five characters of each ZIP code. For example, if one ZIP code contains five characters and the other ZIP code contains nine characters, the comparison ignores the last four characters of the nine-character ZIP code.

If a transaction with a valid address but a blank zip code is received, address verification can still be performed if the POS-ISS-AVS-BLANK-ZIP-CDE-ALWD Logical Network Configuration File (LCONF) param is set to Y.

The following table demonstrates how the interim address verification values are computed.

Information in Message or Database		Interim Address Verification Value	
ZIP Code	Address	ZIP Code	Address
91234	1 Elm Street	91234	1
912340615	123 1st Street	912340615	123
912340615	123 First Street	912340615	123
85282	89 25th Avenue	85282	89
704330123	22 Walnut Street #23	704330123	22
70454	P. O. Box 12345	70454	12345
70454	P. O. Box 123456	70454	12345
WD1 1LA	62 Hilshire Avenue	WD1 1LA	62
HP41JZ	86 5th Street	HP41JZ	86

Address Verification Status Code

The address verification status code serves two purposes. It either identifies the result of comparing the two interim address verification values or identifies the reason that a comparison was not made. The valid values for comparison results are as follows:

A = Address. Addresses matched, but ZIP codes did not match or ZIP code is blank.
N = No. Addresses did not match and ZIP codes did not match.
W = Whole ZIP. Nine-digit ZIP codes matched, but addresses did not match.
X = Exact. Addresses and nine-digit ZIP codes matched.
Y = Yes. Addresses and five-digit ZIP codes matched.
Z = ZIP. Five-digit ZIP codes matched, but addresses did not match.

The following values are also valid, but are used only by Visa. They are not set by the BASE24-pos address verification function:

B = Street address match. The postal code is not verified because of incompatible formats. (Acquirer sends both street address and postal code.)

C = Street address and postal code is not verified because of incompatible formats. (Acquirer sends both street address and postal code.)

D = Street addresses and postal codes match.

G = Global non-participant in the Address Verification Service. Address is not verified for international transactions.

I = Address information is not verified for international transactions.

M = Street address and postal code match.

P = Postal code match. Street is not verified because of incompatible formats. (Acquirer sends both street address and postal code.)

The following table demonstrates possible address verification status codes. Values are arranged according to the quality of the match between the two interim address verification values using the Banknet algorithm.

Information in Message		Information in Database		Verification Status Code	
ZIP Code	Address	ZIP Code	Address		
912340615	1231	912340615	1231	X	Exact: Addresses and nine-digit ZIP codes match

Information in Message		**Information in Database**		**Verification Status Code**	
ZIP Code	**Address**	**ZIP Code**	**Address**		
91234	1	91234	1	Y	Yes: Addresses and five-digit ZIP codes match **Note:** If the ZIP code length is greater than five digits but less than nine digits, the algorithm tries to match only the first five digits of the ZIP code. Therefore, if a seven-digit ZIP code of HP41JAB was matched against HP41JLA, the resulting status code would still be Y.
912340615	1	91234	1		
91234	1	912340615	1		
HP41JZ	865	HP41J	865		
70454	12345	70453	12345	A	Address: Only addresses match
WD1 1LA	62	WD11L	62		
912340615	123	912340615	1231	W	Whole: Only nine-digit ZIP codes match
91234		91234	1	Z	ZIP: Only five-digit ZIP codes match
912340615		91234	1		
91234		912340615	1		
68154	432	68065	234	N	No: Addresses do not match and ZIP codes do not match

Note: A blank address in the message does not match a blank address in the database and a blank ZIP code in the message does not match a blank ZIP code in the database.

BASE24-pos, the VisaNet Interchange, and the Banknet Interchange each have status codes that identify situations when address verification could not be performed. The codes are assigned by the entity that attempts to perform address verification.

If the BASE24-pos Device Handler/Router/Authorization process attempts to perform address verification on a transaction but cannot, one of the BASE24-pos codes defined in the table below is logged to the BASE24-pos log file and will appear on BASE24-pos reports for the transaction.

Value Logged/ Reported by BASE24-pos	Description
E	Error — The transaction was not eligible for address verification or an editing error occurred while attempting to process the message.
R	Retry — Primary and secondary authorizers were unavailable or declined the transaction and address verification was not performed on BASE24-pos.
S	Service not supported — BASE24-pos authorized the transaction, but did not have the add-on Address Verification module.
U	Unavailable — Address information was not available to the processor performing address verification.
blank	Address verification information was not included in this transaction.
0 (zero)	Address verification information was included in this transaction, but was not verified. A transaction to be verified by a host or interchange carries this code. A transaction to be verified on BASE24-pos, but declined before address verification could be performed also carries this code.

If BASE24-pos Interchange Interface processes send the above response codes to the VisaNet or Banknet Interchange, the respective interface process converts the BASE24-pos codes to the equivalent interchange-specific codes. These translated codes are shown in the Value Received columns of the table on the following page.

If an interchange attempts to perform address verification on a transaction but cannot, it sends the interchange-specific code to the BASE24-pos product. BASE24-pos then logs the code sent by the interchange or the translated code equivalent to the BASE24-pos log file. The code then appears in BASE24-pos reports. These codes are shown in the Value Sent columns of the table below.

Value Logged/reported by BASE24-pos	VisaNet		Banknet	
	Value Sent	Value Received	Value Sent	Value Received
E	---	U	---	R
R	R	U	R	R
S	---	U	S	S
U	U	U	U	U
blank	---	blank	---	R
0 (zero)	---	U	---	R

Note: The interchanges cannot send the full range of codes that BASE24-pos is capable of logging. The hyphens (---) in the table above show that a BASE24-pos equivalent value cannot be sent from the interchange. In such cases, the status code fields in the messages sent to BASE24-pos are blank.

Address Verification Timing

Since the address verification status code does not impact authorization results, address verification and transaction authorization do not have to be performed at the same point in the authorization process. Address verification can be performed during processing by the VisaNet Interchange, Banknet Interchange, a host computer, or the BASE24-pos Device Handler/Router/Authorization process. However, the BASE24-pos product does not allow address verification after a transaction is authorized.

Address Verification Alternatives

Address verification can be performed at three possible points during processing:

- Before the BASE24-pos Device Handler module receives the transaction. The VisaNet or Banknet Interchange can perform address verification before it forwards a transaction to the BASE24-pos product. In this case, BASE24-pos retains the address verification status code received from the interchange.
- While the BASE24-pos product is processing the transaction. The BASE24-pos Device Handler/Router/Authorization process can perform address verification if the add-on BASE24-pos Address Verification module has been installed. In this case, the BASE24-pos Device Handler/Router/Authorization process determines the address verification status code and authorizes the transaction or routes it to a host for authorization.
- After BASE24-pos routes the transaction. An interchange or a host computer can perform address verification on a transaction routed to it by BASE24-pos. The BASE24-pos product does not allow address verification after a transaction is authorized, so the transaction must also be authorized by the interchange or host computer.

Flexibility

Address verification is flexible and does not need to be performed at the same point for all transactions. The BASE24-pos Device Handler/Router/Authorization process performs address verification whenever an incoming message contains address and ZIP code information without an address verification status code. For example, if a transaction coming from the VisaNet or Banknet Interchange carries address and ZIP code information with an address verification status code of blank or zero, the BASE24-pos Device Handler/Router/Authorization process determines the appropriate address verification status code or routes the transaction to an interchange or host to perform address verification. The VisaNet or Banknet Interchange can also determine the address verification status for a transaction before the transaction is received by BASE24-pos. In this case, BASE24-pos logs the address verification results it receives from the interchange.

Transaction Destinations

BASE24-pos does not send the information contained in a single address verification transaction to multiple destinations for processing. For example, when BASE24-pos receives a transaction, it does not send the financial information to the host and the address verification information to the interchange. The entire transaction must go to one destination or the other.

This restriction applies only to splitting the information from different fields contained in a single transaction. This restriction should not be confused with screening a transaction on BASE24-pos before sending the transaction to a host. For example, when BASE24-pos receives a transaction, it can perform address verification, set the address verification status code in the message, and then send the entire message to a host for authorization.

BASE24-pos Address Verification Processing

Address verification processing in the BASE24-pos standard product consists of carrying the information in internal and external messages to allow a host or an interchange to verify the address and provide a response. Authorization processing in the BASE24-pos standard product is limited to logging the address verification information to the POS Transaction Log File (PTLF) and Interchange Log Files (ILFs). With the add-on BASE24-pos Address Verification module, the BASE24-pos Device Handler/Router/Authorization process can perform address verification using information maintained in the CAF.

BASE24-pos Address Verification message processing consists of three phases. The processing performed during the first phase—entering the BASE24-pos system—depends on where the message originates. The processing performed during the middle phase—within BASE24-pos processing—depends on whether or not the BASE24-pos Device Handler/Router/Authorization process is verifying the address. The processing performed during the last phase—leaving the BASE24-pos system—depends on where the message is sent for authorization.

Transaction Eligibility

All transactions processed by the BASE24-pos Device Handler/Router/Authorization process and the Banknet Interchange are eligible for address verification. However, the VisaNet Interchange restricts the transaction types that are eligible for address verification to the following:

- Card verification from any merchant
- Mail or telephone order from any merchant
- Regular purchase from an airline merchant

The VisaNet ISO Interface immediately assigns a status code of E (Error) to an incoming transaction that is ineligible and BASE24-pos retains this status code while processing the transaction.

Two VisaNet External Message elements identify eligible transactions:

- Merchant code (element 18)
- POS condition code (element 25)

The following table shows how the VisaNet ISO Interface uses these elements to identify transactions that are eligible for address verification.

Transaction	Merchant Code	POS Condition Code
Card verification from any merchant	Not used	51
Mail or telephone order from any merchant	Not used	08
Regular purchase from an airline merchant	3000–3299, 4511	00

BASE24-pos Standard Internal Message

Four fields in the BASE24-pos Standard Internal Message (PSTM) contain information used during BASE24-pos Address Verification processing. These fields are defined below.

Address (ADDR-FLDS.ADDR) — This field contains the customer address from the external message received by BASE24-pos.

ZIP code (ZIP-CDE) — This field contains the customer ZIP code from the external message received by BASE24-pos.

Address verification status (ADDR-FLDS.ADDR-VRFY-STAT) — This field contains a code describing address verification results.

Address type (ADDR-TYP) — A value of 98 in this field indicates the presence of address verification data in the message. The BASE24-pos process receiving the message sets the value in this field to 98 if the address, ZIP code, or address verification status fields contain data. Any other value in this field indicates the absence of address verification data.

Note: Although these fields occur within the PSTM, the information in these fields is actually token data defined in the Address Verification token (token ID 01). Unlike other tokens, the Address Verification token is not appended to the end of the PSTM.

Entering the BASE24-pos System

Transactions containing address verification data can enter the BASE24-pos system using a number of interfaces, depending on the system configuration. Each of the interfaces in the following list converts an external message into the PSTM and sets the ADDR-TYP field value to 98 if address verification data is present.

- Merchant Host Interface (MHI). No configuration changes are necessary for BASE24-pos to accept MHI transactions containing address verification data. However, address verification field IDs (FIDs) must be set in the MHI Configuration File (MCNF) before information can be returned in MHI responses. FID Z contains the address verification status. Optionally, FID A and FID Y can be set to echo the address and ZIP code values included in the original transaction request.
- VisaNet ISO Interface. No configuration changes are necessary for BASE24-pos to accept transactions containing address verification data from the VisaNet Interchange. The VisaNet ISO Interface performs the eligibility test described earlier and sets the address verification status code if necessary.
- Banknet ISO Interface. The Banknet ISO Interface requires configuration changes to the Banknet Institution Identification File (BNIIF) before BASE24-pos can receive address verification data from the Banknet Interchange. This configuration change must be made before logging on to the Banknet Interchange.
- ANSI Host Interface. The ANSI Host Interface requires a configuration change in the Host Configuration File (HCF) before it can accept address verification data. This HCF setting is required to implement an extended format of the BASE24-pos Standard External Message (PSEM).
- ISO Host Interface. The ISO Host Interface may require configuration changes to the External Message File (EMF) before it can accept address verification data. EMF settings determine which message elements are included in messages sent between BASE24-pos and a host. Elements P-44 and P-63 in the ISO external message are defined for address verification data in BASE24-pos messages. Element S-127 is defined for address verification data in BASE24-from host maintenance messages.

  The standard EMF defaults include these elements. Therefore, no configuration changes are necessary for these elements if EMF default values are used instead of EMF records and if address verification is implemented. If EMF records are used, these elements must be added to the EMF record before BASE24-pos can accept address verification data. If the default settings are used, but address verification is not to be implemented, EMF records can be created to override the standard address verification defaults.

- BIC ANSI Interface. The BIC ANSI Interface requires a configuration change in the Logical Network Configuration File (LCONF) before it can accept address verification data. An LCONF setting determines which byte map value and external message format is used for BASE24-pos messages. ANSI external message bytes 11 and 12 carry address verification data and are defined in the BASE24-pos messages and byte maps.
- BIC ISO Interface. The BIC ISO Interface may require configuration changes in the EMF before it can support address verification data. EMF settings determine which message elements are included in messages sent between BASE24-pos and a co-network. Elements P-44 and P-63 in the ISO external message are defined for address verification data in BASE24-pos messages.

  The standard EMF defaults include these elements. Therefore, no configuration changes are necessary for these elements if EMF default values are used instead of EMF records and if address verification is implemented. If EMF records are used, these elements must be added to EMF records before the BIC ISO Interface can accept address verification data. If the default settings are used, but address verification is not to be implemented, EMF records can be created to override the standard address verification defaults.

Within BASE24-pos Processing

Without the add-on BASE24-pos Address Verification module, processing is limited to logging available address verification information to the PTLF and ILFs.

With the add-on module, the BASE24-pos Device Handler/Router/Authorization process can perform address verification on any transaction it authorizes. The add-on module can also perform address verification while screening transactions bound for a host for authorization. In both situations, the Device Handler/Router/Authorization process performs address verification as the first step after locating the cardholder’s CAF record. The value in the ADDRESS VERIFICATION ALGO field in the Card Prefix File (CPF) specifies the algorithm the Device Handler/Router/Authorization process uses to verify an address. The value in the CHECK IF HOST IS ONLINE—ADDR VERIFY field in the CPF specifies whether the Device Handler/Router/Authorization process performs address verification while screening host-bound transactions.

The Device Handler/Router/Authorization process does not modify the result of address verification that has already been performed. The Device Handler/Router/Authorization process checks the address verification status code in the PSTM and only performs address verification if this value is blank or zero.

Leaving the BASE24-pos System

Transactions can leave the BASE24-pos system using a number of interfaces, depending on the system configuration. Since these transactions can be processed with or without address verification information, BASE24-pos uses a value of 98 in the ADDR-TYP field of the PSTM to identify the presence of address verification data. If address verification data is present in the PSTM, each of the interfaces in the following list can include or discard the data, depending on whether the destination host or interchange supports address verification. Data from the address, ZIP code, and address verification status fields in the PSTM is transferred to external message fields, but the ADDR-TYP field data in the PSTM is not.

- VisaNet ISO Interface. No configuration changes are necessary for address verification data to leave BASE24-pos. If a card verification transaction containing address verification data leaves BASE24-pos for the VisaNet Interchange, the VisaNet ISO Interface sets element P-25 in the VisaNet External Message to 51 (address verification only). For transactions other than card verification, the condition code value in the POS-COND-CDE field in the PSTM is moved to element P-25 in the external message.
- Banknet ISO Interface. The Banknet ISO Interface requires configuration changes to the Banknet Institution Identification File (BNIIF) before the Interchange Interface process can send address verification data to the Banknet Interchange. This configuration change must be made before logging on to the Banknet Interchange.
- ANSI Host Interface. The ANSI Host Interface requires a configuration change in the HCF before it can send address verification data. If a transaction contains address verification data in the PSTM, but the HCF configuration setting indicates not to include address verification, the ANSI Host Interface does not move any address verification data from the PSTM to the PSEM.
- ISO Host Interface. The ISO Host Interface may require configuration changes to the EMF and the Token File (TKN) before it can send address verification data. EMF settings determine which message elements are included in messages sent between BASE24-pos and a host. The TKN

defines which token information is logged to and extracted from the POS Transaction Log File (PTLF) and which token information is sent in the external message.

No configuration changes are necessary if EMF default values are used instead of EMF records. EMF settings determine which message elements are included in messages sent between BASE24-pos and a host. Elements P-44 and P-63 in the ISO external message are defined for address verification data in BASE24-pos messages.

The standard EMF defaults indicate that address verification data be included in the message. Therefore, no configuration changes are necessary for these elements if EMF default values are used instead of EMF records. If EMF records are used, these elements must be added to EMF records before BASE24-pos can send address verification data. If the default settings are used, but address verification is not to be implemented, EMF records can be created to override the standard address verification defaults.

The TKN must be configured to include the Address Verification Token (token ID 01) if address verification is implemented. This token information must be added before BASE24-pos can transfer address verification data from the PSTM to the external message.

For example, if a transaction contains address verification data in the PSTM, but the EMF P-44 and P-63 settings are blank and the Address Verification Token is not defined in the TKN, the ISO Host Interface does not move any address verification data from the PSTM to the ISO external message.

- BIC ANSI Interface. The BIC ANSI Interface requires a configuration change in the LCONF before it can send address verification data. If a transaction contains address verification data in the PSTM, but the LCONF configuration setting indicates not to include address verification, the BIC ANSI Interface does not move any address verification data from the PSTM to the ANSI external message.
- BIC ISO Interface. The BIC ISO Interface may require configuration changes to existing EMF records and TKN records before it can send address verification data. EMF settings determine which message elements are included in messages sent between BASE24-pos and a co-network. The TKN defines which token information is logged to and extracted from the Interchange Log File (ILF) and which token information is sent in the external message.

  Elements P-44 and P-63 in the ISO external message are defined for address verification data in BASE24-pos messages. The standard EMF defaults include these elements. Therefore, no configuration changes are necessary for these elements if EMF default values are used instead of EMF records and if address verification is implemented. If EMF records are used, these elements must be added to EMF records before the BIC ISO Interface can send address verification data. If the default settings are used, but address verification is not to be implemented, EMF records can be created to override the standard address verification defaults.

The TKN must be configured to include the Address Verification Token (token ID 01) if address verification is implemented. This token information must be added before BASE24-pos can transfer address verification data from the PSTM to the external message.

For example, if a transaction contains address verification data in the PSTM, but the EMF P-44 and P-63 settings indicate not to include address verification and the Address Verification Token is not defined in the TKN, the BIC ISO Interface does not move any address verification data from the PSTM to the ISO external message.

BASE24-pos Address Verification Reporting

BASE24-pos reports and log files include address verification information for inquiry and reference purposes. The Daily Cardholder Activity Report (B24POS02) and Daily Retailer Activity Report (B24POS12) contain three additional fields for each transaction involving address verification—address, ZIP code, and status code. The address and ZIP code on the reports are the ones received with the transaction. The status code on the reports is the final result of address verification processing.

If configured to do so in the Token File (TKN), the address verification address, ZIP code, and status code information is logged to the POS Transaction Log File (PTLF). This information is available for online perusal of transactions using the PTLF perusal function. Refer to the ***BASE24-pos Files Maintenance Manual*** for information about perusing the PTLF.

Interchange Log File (ILF) records written by the VisaNet ISO Interface, the Banknet ISO Interface, and the BIC ISO or ANSI Interface include the address verification status code with transactions.

The Online Maintenance File (OMF) includes the address verification address and ZIP code fields in its records of CAF additions, deletions, and updates performed using BASE24 CRT access.

The Update Log File (ULF) includes the address verification address and ZIP code fields in its records of CAF additions, deletions, and updates performed using the BASE24-from host maintenance product.

The PTLF, ILF, OMF, and ULF are extractable in the disk format using the BASE24 Super Extract process (i.e., the binary fields are not converted to ASCII). The PTLF and ILF also are extractable in ASCII-only format (i.e., the binary fields are converted to ASCII).

Impacts on Other Products

BASE24-pos Address Verification processing impacts the BASE24 Refresh process and the From Host Maintenance process. The impacts to each of these processes are summarized below.

- The BASE24 Refresh process can be used to maintain the address and ZIP code fields in the Address Verification segment of the CAF when the add-on BASE24-pos Address Verification module is installed.
- The From Host Maintenance process can be used to access the address and ZIP code fields in the Address Verification segment of the CAF when the add-on BASE24-pos Address Verification module is installed.

2: BASE24-pos Implementation

This section provides information for implementing address verification in a BASE24-pos system. Information is included for the standard BASE24-pos product and the BASE24-pos add-on Address Verification module. The following aspects of implementing Address Verification are presented in this section:

- Setting up a host
- Setting up a BIC ISO or ANSI Interface
- Setting up a Banknet ISO Interface
- Setting up a VisaNet ISO Interface
- Setting up a card issuer
- Maintaining cardholder addresses on BASE24
- Maintaining cardholder addresses at an interchange
- Perusing cardholder transactions

Introduction

BASE24-pos Address Verification processing is controlled by configuration file settings. These settings specify whether a host or interchange supports address verification information in the messages it exchanges with BASE24-pos. If the BASE24-pos add-on Address Verification module is being used, card issuer and cardholder information files must also be maintained.

The procedures in this section are organized according to the type of information being updated. Host configuration includes the BASE24-from host maintenance product. Since all of this information is contained in existing BASE24 files, the files maintenance procedures provided in this section assume an existing record is being updated.

The key used to display screen 1 of a file depends on the screen or menu currently displayed. When the Logon screen, Virtual Menu, or a product menu is displayed, the **F1** key must be pressed after completing the FILE DESTINATION field. When any other screen or menu is displayed, the **F16** key must be pressed after completing the FILE DESTINATION field. The files maintenance procedures provided in this section assume the Logon screen, Virtual Menu, or a product menu is displayed. The **F16** key can be substituted for the **F1** key when necessary.

Setting Up a Host

Address verification data is not always automatically included in messages exchanged between BASE24-pos and host computers. In some cases, messages moving between BASE24-pos and a particular host computer must be modified before they can carry the additional data.

> **Warning:** Address verification data should be added to the messages between BASE24-pos and a host computer only after the host computer has been reprogrammed to process the additional data. The procedures presented here should be performed in cooperation with host computer center personnel.

The first step in setting up a host is identifying the message format used to communicate between that host and BASE24-pos. Two message formats are used by BASE24-pos—one based on standards set by the American National Standards Institute (ANSI) and one based on standards set by the International Organization for Standardization (ISO)—and the message format determines the steps that must be taken to include address verification data in the message. Host computer center personnel can also assist in identifying which message format is being used to communicate with their system.

Modifying the ANSI Message Format

BASE24-pos uses an extended version of the POS External Message (PSEM) to carry address verification data. The value in the PSEM TYPE field on Host Configuration File (HCF) screen 7 determines whether the PSEM format in use is the standard version or the extended version. This value also determines whether the PSEM in use includes only authorization fields, only financial fields, or authorization and financial fields. For address verification, it is necessary to change the PSEM TYPE field entry from its current standard version value to the comparable extended version value, as shown in the table below.

If the current PSEM TYPE value is:	Change the PSEM TYPE value to:
0 (PSEMA and PSEMF)	3 (PSEMAE and PSEMFE)
1 (PSEMA)	4 (PSEMAE)
2 (PSEMF)	5 (PSEMFE)

To change the PSEM TYPE field value, proceed as follows:

1. Obtain the Data Processing Center (DPC) number and BASE24 Host Interface symbolic name that identify the DPC adding address verification information to its messages.
2. Display HCF screen 1 by entering HCF in the FILE DESTINATION field at the bottom of the CRT access screen and pressing the **F1** key.
3. Read the HCF record being changed by completing the DPC NUMBER and HISF NAME fields and pressing the **F2** key.
4. Display HCF screen 7 by entering 7 in the NEW PAGE field at the bottom of the CRT access screen and pressing the **F7** key.
5. Change the PSEM TYPE value according to the table shown above.
6. Update the HCF record by pressing the **F5** key.

HCF Screen 7

HCF screen 7 is shown below, with a description of the PSEM TYPE field on the following page.

```
BASE24-POS   HOST CONFIGURATION       LLLL        YY/MM/DD  HH:MM  07 OF 25
                  DPC NUMBER: 0000        HISF NAME: P1A^HISF1

                           POS PRODUCT DATA

                       PSEM TYPE: 3   (BOTH AUTH AND FINANCIAL FORMS)
                    AUTH PROCESS:
                  MESSAGE FORMAT: 01  (VARIABLE FORMAT)
               RELEASE INDICATOR: 01  (CURRENT RELEASE)
           REFERRAL PHONE NUMBER:

                            TIMER LIMITS

                  OUTBOUND LIMIT:   15 (SEC)
                   INBOUND LIMIT:   15 (SEC)
                      COMPLETION:   30 (SEC)
                  COMPLETION ACK:   30 (SEC)
                  QUEUE SUBTRACT:    5 (SEC)


*********************************** BASE24 ***********************************
 NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                          F12-HELP
```

PSEM TYPE — A code describing the type of POS External Message (PSEM) format being used. Extended formats, values 3 through 5, allow additional space for address verification information. If address verification information is not being exchanged with a particular host, the standard formats, values 0 through 2, should be used. This field is used for an ANSI host only. Valid values are as follows:

0 = Both Authorization and Financial forms (PSEMA and PSEMF)
1 = Authorization form only (PSEMA)
2 = Financial form only (PSEMF)
3 = Both Authorization and Financial forms (PSEMAE and PSEMFE)
4 = Authorization form only (PSEMAE)
5 = Financial form only (PSEMFE)

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0 (BOTH AUTH AND FINANCIAL FORMS)

Modifying the ISO Message Format

BASE24-pos uses two elements in the ISO external message to carry address verification data to a host. Element P-44 in the ISO external message contains the address verification status, and element P-63 contains the address and ZIP code information in the Address Verification Token. Element S-127 contains the address and ZIP code information for the BASE24-from host maintenance product.

The External Message File (EMF) default settings for elements P-44, P-63, and S-127 enable the host to exchange address verification data with the BASE24-pos and BASE24-from host maintenance products. When implementing address verification and using default settings in the EMF, no configuration changes are necessary. The default settings are shown in the table on the following page.

PROD # Field:	MSG TYP Field:	Default Setting: Conditional (C)
02 (POS)	0200	P-44 and P-63
02 (POS)	0210	P-44 and P-63
02 (POS)	0220	P-44 and P-63
08 (FHM)	0300	S-127
08 (FHM)	0310	S-127

When address verification is **not** to be implemented but the standard EMF default setting for element P-44 is used, EMF records can be created for POS 0200, 0210, and 0220 messages to override the default setting for that element. These EMF records must contain a blank instead of a C for element P-44 being omitted.

Note: If token data other than the Address Verification Token is to be sent in the external message, element P-63 must contain a C; P-63 cannot contain a blank because other token data can also be carried in that element.

If address verification is to be implemented and EMF records are used, records for POS 0200, 0210, and 0220 messages must be added or modified to enable the exchange of address verification data. These records must contain the standard default settings for P-44 and P-63 as shown in the default table above. Existing elements for each message type should not be changed.

Records for BASE24-from host maintenance 0300 and 0310 messages also must be changed in the EMF to match the default value for element S-127 in the table above for each ISO host using the BASE24-from host maintenance product to maintain CAF address verification data.

Note: If the BASE24-from host maintenance product is not used, procedures for updating the EMF records for BASE24-from host maintenance message types can be omitted.

> **Warning:** Special care must be used when updating the EMF. Access to EMF screens 1, 2, and 3 is needed in order to perform this files maintenance operation. EMF screen 3 is used to assign IMS or CICS transaction code equivalents to BASE24 transaction codes and can consist of up to five pages. The last page containing transaction codes must be displayed when the **F5** key is pressed to update the record or transaction codes will be lost. For example, if the **F5** key is pressed while screen 1 or 2 is displayed, all transaction codes on the pages of screen 3 will be lost. If the **F5** key is pressed while a page of screen 3 is displayed, but it is not the last page containing transaction codes, the transaction codes on the remaining pages of screen 3 will be lost.

To change the EMF records, proceed as follows:

1. Obtain the DPC number and BASE24 Host Interface symbolic name that identify the DPC adding address verification information to its messages.
2. Obtain the symbolic name that identifies the From Host Maintenance process for the DPC adding address verification information to its messages.
3. Display EMF screen 1 by entering EMF in the FILE DESTINATION field at the bottom of the CRT access screen and pressing the **F1** key.
4. Read the EMF record being changed by completing the field information shown below and pressing the **F2** key. Each column represents a separate EMF record. If no record is found, the Host Interface or From Host Maintenance process is using EMF default values.

INTERF TYP	HOST			FHM	
DPC/MOD #	DPC number assigned to the host			0	0
PROCESS NAM	Symbolic name of Host Interface process			Symbolic name of From Host Maintenance process	
PROD #	02	02	02	08	08
MSG TYP	0200	0210	0220	0300	0310
IN-OUT-IND	B	B	B	I	O

5. Enter a C in the field directly to the right of the element numbers shown in the table, but do **not** press the **F5** key.
6. Display EMF screen 3 by pressing the **F9** key twice or by entering a 3 in the NEW PAGE field at the bottom of the CRT access screen and pressing the **F7** key.
7. If the value in the NUMBER OF TRAN CODES field on EMF screen 3 is 000, update the EMF record by pressing the **F5** key and continue with step 9.
8. If the value in the NUMBER OF TRAN CODES field on EMF screen 3 is not 000, display the last EMF screen 3 page containing transaction code entries by pressing the **F8** key before updating the EMF record. Each EMF screen 3 page contains 30 transaction codes, and the range of transaction codes being displayed on a page is shown on the same line as the NUMBER OF TRAN CODES field. Display the next page until the NUMBER OF TRAN CODES field value falls within the range shown, then update the EMF record by pressing the **F5** key.
9. Return to EMF screen 1 by pressing the **F11** key. Repeat steps 4 through 9 for each of the remaining message types.

EMF screens 1 and 3 are displayed below and on the following page. The NUMBER OF TRAN CODES field value on screen 3 is 002 and the range of transaction codes being displayed on the page is 1–30. Since 2 is between 1 and 30, this EMF record can be updated without losing any transaction codes.

Note: EMF screen 2 allows institutions to specify which external message fields are used when computing message authentication code (MAC) values. It does not impact the BASE24-pos Address Verification product. Refer to the ***BASE24 Base Files Maintenance Manual*** for more information about this screen.

EMF Screen 1

EMF screen 1 is shown below with elements P-44, P-63, and S-127 set to conditional.

```
BASE24-BASE  EXT MESSAGE FILE         LLLL        YY/MM/DD  HH:MM  01 OF 03
INTERF TYP:      DPC/MOD #:    0 PROCESS NAM:                  PROD #: 00 BASE
MSG TYP: 0000   IN-OUT-IND:           TOKEN GROUP:       FULL MSG MAC: N (NO)
  P-1      P-17     P-33     P-49     S-65     S-81     S-97     S-113
  P-2      P-18     P-34     P-50     S-66     S-82     S-98     S-114
  P-3      P-19     P-35     P-51     S-67     S-83     S-99     S-115
  P-4      P-20     P-36     P-52     S-68     S-84     S-100    S-116
  P-5      P-21     P-37     P-53     S-69     S-85     S-101    S-117
  P-6      P-22     P-38     P-54     S-70     S-86     S-102    S-118
  P-7      P-23     P-39     P-55     S-71     S-87     S-103    S-119
  P-8      P-24     P-40     P-56     S-72     S-88     S-104    S-120
  P-9      P-25     P-41     P-57     S-73     S-89     S-105    S-121
  P-10     P-26     P-42     P-58     S-74     S-90     S-106    S-122
  P-11     P-27     P-43     P-59     S-75     S-91     S-107    S-123
  P-12     P-28     P-44 C   P-60     S-76     S-92     S-108    S-124
  P-13     P-29     P-45     P-61     S-77     S-93     S-109    S-125
  P-14     P-30     P-46     P-62     S-78     S-94     S-110    S-126
  P-15     P-31     P-47     P-63 C   S-79     S-95     S-111    S-127 C
  P-16     P-32     P-48     P-64     S-80     S-96     S-112    S-128

*********************************** BASE24 ***********************************
 NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
F7-DEFAULTS  F12-HELP  IF THERE ARE T-CODES DON'T ADD/UPDATE FROM THIS SCREEN
```

EMF Screen 3

EMF screen 3 is shown below with two IMS transaction code equivalent entries.

```
BASE24-BASE  EXT MESSAGE FILE         LLLL        YY/MM/DD  HH:MM  03 OF 03
INTERF TYP:      DPC/MOD #:    0 PROCESS NAM:                 PROD #: 00 BASE
MSG TYP: 0000   IN-OUT-IND:           TOKEN GROUP:       FULL MSG MAC: N (NO)

NUMBER OF TRAN CODES: 002   THIS SCREEN IS FOR TRAN CODES 1 - 30
NOTE: ADD/UPDATE FROM THE LAST SCREEN FULL OF T CODES TO BE INCLUDED
ROW TRAN    IMS TRAN   L     TRAN    IMS TRAN   L     TRAN    IMS TRAN   L

 1  100100  W0000001   8
 2  101100  W0000002   8
 3
 4
 5
 6
 7
 8
 9
10


********************************** BASE24 **********************************
 NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
F8 - GET-THE-SCREEN-AGAIN-UP-TO-5-TIMES      F12 -HELP
```

Setting Up the Token File (TKN) for an ISO Host

Address verification data is carried in the Address Verification Token (token ID 01). In order for this data to be logged to the POS Transaction Log File (PTLF), extracted from the PTLF, or sent to the host in the ISO-based external message, the Token File (TKN) must be configured with the Address Verification Token.

TKN screen 2 contains fields specifying which tokens are logged to the PTLF. TKN screen 3 contains fields specifying which tokens are extracted from the PTLF. TKN screen 4 contains fields specifying which tokens are sent in the external message.

Refer to the ***BASE24 Tokens Manual*** for procedures to be used when adding or updating records in the TKN for BASE24-pos Address Verification.

Setting Up a BIC Interface

The BASE24 Interchange (BIC) Interface can include address verification information in its messages. However, address verification data is not always automatically included in messages passed between BIC Interchanges. In some cases, these messages must be modified before they can carry the additional data.

> **Warning:** Address verification data should be added to the messages between the BIC Interface and a co-network only after the co-network is prepared to process the additional data. The procedures presented here should be performed in cooperation with data processing operations personnel at the other site.

Like setting up a host, the first step in setting up a BIC Interface is identifying the message format used for communication. Once again, the ANSI and ISO message formats can be used, and the message format determines the steps that must be taken to include address verification data in the message.

Modifying the BIC ANSI Message Format

The BIC ANSI Interface adds elements 11 and 12 to the ANSI external message to carry address verification data. The BIC-VERSION-NUM param value in the Logical Network Configuration File (LCONF) determines whether the ANSI external message format used with the co-network should include the address verification elements.

Use procedures presented in sections 2, 3, and 5 of the ***BASE24 Logical Network Configuration File Manual*** to change the BIC-VERSION-NUM param value from a code indicating release 5.3 without address verification to a code indicating release 5.3 with address verification, as shown in the table on the following page. If the code indicates a release other than 5.0, the address verification option is not valid.

A BIC-VERSION-NUM param of:	Indicates that:
00000100 (Previous release)	Address verification cannot be supported on this software release. The co-network must upgrade to the current software release.
00000200 (Current release)	The co-network is on the current software release, but does not want to process messages containing address verification data.
00000300 (Address verification)	The co-network can process messages including address verification data.

The ANSI BIC Interface checks the BIC-VERSION-NUM param values in each network during logon. If the values do not match, logon is not allowed.

Modifying the BIC ISO Message Format

The BIC ISO Interface uses two elements in the ISO external message to carry address verification data. Element P-44 in the ISO external message contains the address verification status code, and element P-63 contains the address and ZIP code information in the Address Verification Token.

The default settings for elements P-44 and P-63 enable BIC ISO Interchanges to exchange address verification data. When implementing address verification and using default EMF settings, no configuration changes are necessary. The default settings are shown in the table below.

MSG TYP Field:	Default Setting: Conditional (C)
0200	P-44 and P-63
0210	P-44 and P-63
0220	P-44 and P-63

When address verification is **not** implemented but the standard EMF default setting for element P-44 is used, EMF records can be created for POS 0200, 0210, and 0220 messages to override the default setting for that element. These EMF records must contain a blank instead of a value of C for element P-44.

Note: If token data other than the Address Verification Token is to be sent in the external message, element P-63 must contain a value of C; P-63 cannot contain a blank because other token data can also be carried in that element.

If address verification is to be implemented and EMF records are used, records for POS 0200, 0210, and 0220 messages must be added or modified to enable the exchange of address verification data. These records must contain the standard default settings for P-44 and P-63 as shown in the default table above. Existing elements for each message type should not be changed.

Warning: Special care must be used when updating the EMF. Access to EMF screens 1, 2, and 3 is needed in order to perform this files maintenance operation. EMF screen 3 is used to assign IMS or CICS transaction code equivalents to BASE24 transaction codes and it can consist of up to five pages. The last page containing transaction codes must be displayed when the **F5** key is pressed to update the record or transaction codes will be lost. For example, if the **F5** key is pressed while screen 1 is displayed, all transaction codes on the pages of screen 3 will be lost. If the **F5** key is pressed while one of the screen 3 pages is displayed, but it is not the last page containing transaction codes, the transaction codes on the remaining pages of screen 3 will be lost.

Three BASE24-pos EMF records must be changed if EMF default values are not used and address verification is to be implemented. To change the EMF records, proceed as follows:

1. Obtain the BIC ISO Interface symbolic name that identifies the interchange adding address verification information to its messages.
2. Display EMF screen 1 by entering EMF in the FILE DESTINATION field at the bottom of the CRT access screen and pressing the **F1** key.

3. Read the EMF record being changed by completing the field information shown below and pressing the **F2** key. Each column represents a separate EMF record. If no record is found, the BIC Interface is using EMF default values.

INTERF TYP	BIC	BIC	BIC
DPC/MOD #	0	0	0
PROCESS NAM	Symbolic name of BIC process		
PROD #	02	02	02
MSG TYP	0200	0210	0220
IN-OUT-IND	B	B	B

4. Enter a C in the field directly to the right of the element numbers shown in the table on the previous page, but do **not** press the **F5** key.

5. Display EMF screen 3 by pressing the **F9** key twice or by entering a 3 in the NEW PAGE field on the bottom of the CRT access screen and pressing the **F7** key.

6. If the value in the NUMBER OF TRAN CODES field on EMF screen 3 is 000, update the EMF record by pressing the **F5** key and continue with step 8.

7. If the value in the NUMBER OF TRAN CODES field on EMF screen 3 is not 000, display the last EMF screen 3 page containing transaction code entries by pressing the **F8** key before updating the EMF record. Each EMF screen 3 page contains 30 transaction codes, and the range of transaction codes being displayed on a page is shown on the same line as the NUMBER OF TRAN CODES field. Display the next page until the NUMBER OF TRAN CODES field value falls within the range shown, then update the EMF record by pressing the **F5** key.

8. Return to EMF screen 1 by pressing the **F11** key. Repeat steps 3 through 8 for each of the remaining message types.

EMF screens 1 and 3 are shown on the following pages. The NUMBER OF TRAN CODES field value is 000; this EMF record can be updated because there are no transaction codes.

Note: EMF screen 2 allows institutions to specify which external message fields are used when computing message authentication code (MAC) values. It does not impact the BASE24-pos Address Verification product. Refer to the ***BASE24 Base Files Maintenance Manual*** for more information about this screen.

EMF Screen 1

EMF screen 1 is shown below with elements P-44 and P-63 set to conditional.

```
BASE24-BASE  EXT MESSAGE FILE         LLLL        YY/MM/DD  HH:MM  01 OF 03
INTERF TYP:      DPC/MOD #:    0 PROCESS NAM:                 PROD #: 00 BASE
MSG TYP: 0000   IN-OUT-IND:          TOKEN GROUP:       FULL MSG MAC: N (NO)
  P-1      P-17     P-33     P-49     S-65     S-81     S-97     S-113
  P-2      P-18     P-34     P-50     S-66     S-82     S-98     S-114
  P-3      P-19     P-35     P-51     S-67     S-83     S-99     S-115
  P-4      P-20     P-36     P-52     S-68     S-84     S-100    S-116
  P-5      P-21     P-37     P-53     S-69     S-85     S-101    S-117
  P-6      P-22     P-38     P-54     S-70     S-86     S-102    S-118
  P-7      P-23     P-39     P-55     S-71     S-87     S-103    S-119
  P-8      P-24     P-40     P-56     S-72     S-88     S-104    S-120
  P-9      P-25     P-41     P-57     S-73     S-89     S-105    S-121
  P-10     P-26     P-42     P-58     S-74     S-90     S-106    S-122
  P-11     P-27     P-43     P-59     S-75     S-91     S-107    S-123
  P-12     P-28     P-44 C   P-60     S-76     S-92     S-108    S-124
  P-13     P-29     P-45     P-61     S-77     S-93     S-109    S-125
  P-14     P-30     P-46     P-62     S-78     S-94     S-110    S-126
  P-15     P-31     P-47     P-63 C   S-79     S-95     S-111    S-127
  P-16     P-32     P-48     P-64     S-80     S-96     S-112    S-128

*********************************** BASE24 ***********************************
 NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
F7-DEFAULTS  F12-HELP  IF THERE ARE T-CODES DON'T ADD/UPDATE FROM THIS SCREEN
```

EMF Screen 3

EMF screen 3 is shown below with no transaction code entries.

```
BASE24-BASE  EXT MESSAGE FILE         LLLL        YY/MM/DD  HH:MM  03 OF 03
INTERF TYP:      DPC/MOD #:    0 PROCESS NAM:                  PROD #: 00  BASE
MSG TYP: 0000   IN-OUT-IND:           TOKEN GROUP:       FULL MSG MAC: N (NO)

NUMBER OF TRAN CODES: 000   THIS SCREEN IS FOR TRAN CODES 1 - 30
NOTE: ADD/UPDATE FROM THE LAST SCREEN FULL OF T CODES TO BE INCLUDED
ROW TRAN    IMS TRAN   L     TRAN    IMS TRAN   L     TRAN    IMS TRAN   L

 1
 2
 3
 4
 5
 6
 7
 8
 9
10


********************************** BASE24 **********************************
 NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
F8 - GET-THE-SCREEN-AGAIN-UP-TO-5-TIMES      F12 -HELP
```

Setting Up the Token File (TKN) for the BIC ISO Interface

Address verification data is carried in the Address Verification Token (token ID 01). In order for this data to be logged to the Interchange Log File (ILF), extracted from the ILF, or sent to the BIC ISO interchange in the ISO-based external message, the Token File (TKN) must be configured with the Address Verification Token.

TKN screen 2 contains fields specifying which tokens are logged to the ILF. TKN screen 3 contains fields specifying which tokens are extracted from the ILF. TKN screen 4 contains fields specifying which tokens are sent in the external message.

Refer to the ***BASE24 Tokens Manual*** for procedures to be used when adding or updating records in the TKN for BASE24-pos Address Verification.

Setting Up the Banknet ISO Interface

The Banknet ISO Interface can also include address verification information in its messages. However, it requires files maintenance in the Banknet Institution Identification File (BNIIF) in order to carry the additional data. The BNIIF is accessed using the Interchange Configuration File (ICF).

BASE24-pos defines processing parameters for the Banknet ISO Interface in the BNIIF. The AVS TYPE field on screen 12 of the BNIIF becomes functional when address verification is implemented. The modifications described in this section are necessary if an institution is a card issuer and is using the Banknet Interchange.

The AVS TYPE field specifies the type of address verification service that Banknet and the issuer use when exchanging address verification data. The type of service must be configured by financial institution group when logging on to the Banknet Interchange. The information in the AVS TYPE field is passed only between the Banknet ISO Interface and the Banknet Interchange in Network Management Messages.

The AVS TYPE field on screen 12 of the BNIIF must be set whether the BASE24-pos Device Handler/Router/Authorization process performs address verification and also authorizes the transaction or just performs address verification and passes the transaction on to the host or interchange for authorization.

To set the AVS TYPE field, proceed as follows:

1. Display ICF screen 1 by entering ICF in the FILE DESTINATION field at the bottom of the CRT access screen and pressing the **F1** key.
2. Read the ICF record being changed by completing the INTERCHANGE FIID and the PROCESS fields and pressing the **F2** key.
3. Display ICF screen 12 by entering 12 in the NEW PAGE field at the bottom of the CRT access screen and pressing the **F7** key.
4. Access the Banknet menu by pressing the **F8** key.
5. Select BNIIF from the Banknet function menu by placing the cursor next to the BNIIF entry and pressing the **F1** key.
6. Read the BNIIF record being changed by completing the FINANCIAL ID field and pressing the **F2** key.
7. Set the AVS TYPE field by entering the appropriate value.
8. Update the BNIIF by pressing the **F5** key.

BNIIF Screen

The BNIIF screen is shown below, with a description of its AVS TYPE field following the screen.

```
BASE24-SWI  BNIIF FILE MAINTENANCE    LLLL       YY/MM/DD  HH:MM  01 OF 01

     FINANCIAL ID:  ____      INTERCHANGE FIID:  ____
   BANKNET MCI ID:  ______       SECURITY CODE:  _____

            SERVICE TYPE: _ ( D/C/B )    TRAFFIC TYPE: _ ( A/I/B )
                          AVS TYPE _ ( 0/1/2/3/4/5 )

BANKNET ISSUER:
           PREFIX    LGTH                 PREFIX    LGTH

        ___________   __               ___________   __
        ___________   __               ___________   __
        ___________   __               ___________   __
        ___________   __               ___________   __
        ___________   __               ___________   __
        ___________   __               ___________   __
        ___________   __               ___________   __
        ___________   __               ___________   __
        ___________   __               ___________   __

********************************** BASE24 *********************************
NEW PAGE:       FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                                   F12-HELP
```

AVS TYPE — A code indicating the amount of information passed to the card issuer and whether Banknet or the BASE24-pos Device Handler/Router/Authorization process performs address verification. Valid values are as follows:

0 = Address verification processing is not supported. This value is valid only for acquirer-only institutions.
1 = BASE24-pos receives complete address data and verifies the address.
2 = BASE24-pos receives condensed address data and verifies the address.
3 = BASE24-pos receives complete address data and Banknet verifies the address.
4 = BASE24-pos receives condensed address data and Banknet verifies the address.
5 = BASE24-pos does not receive any address data and Banknet verifies the address.

Address verification data can be exchanged in either complete or condensed form. Complete address data includes the entire address and zip code. Condensed address data includes only the first five digits of the address. For example, if the given address were 1234 North 1st Street, the condensed address data would be 12341.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	No default value

Setting Up the VisaNet ISO Interface

The VisaNet ISO Interface includes address verification information in its messages without any BASE24 files maintenance or configuration changes.

Setting Up a Card Issuer

BASE24-pos defines processing parameters for each card issuer in the Card Prefix File (CPF). Two CPF fields become functional when the add-on BASE24-pos Address Verification module has been installed.

The ADDRESS VERIFICATION ALGO field on screen 7 specifies the algorithm used when the BASE24-pos Authorization module performs address verification involving cards with this prefix. This field must be set whether the Authorization module performs address verification and also authorizes the transaction or just performs address verification and passes the transaction on to a host for authorization.

The CHECK IF HOST IS ONLINE ADDR VERIFY field, also on screen 7, specifies whether the BASE24-pos Authorization module should perform address verification on transactions involving cards with this prefix before sending them to a host for authorization.

Entries are required in both of these fields. However, neither of these fields affect processing unless the add-on BASE24-pos Address Verification module has been installed.

Modifying Card Issuer Information

The CPF contains card issuer information for address verification. To set the CPF fields, proceed as follows:

1. Display CPF screen 1 by entering CPF in the FILE DESTINATION field at the bottom of the CRT access screen and pressing the **F1** key.
2. Read the CPF record being changed by completing the PREFIX, PAN LENGTH, and FIID fields and pressing the **F2** key.
3. Display CPF screen 7 by entering 7 in the NEW PAGE field at the bottom of the CRT access screen and pressing the **F7** key.
4. Set the ADDRESS VERIFICATION ALGO field to either a V or an M. A setting of V indicates that the VisaNet algorithm is used when the BASE24-pos Authorization module performs address verification. A setting of M indicates that the Banknet algorithm is used when the BASE24-pos Authorization module performs address verification.

5. If the BASE24-pos Authorization module is to perform address verification before sending the transaction to a host for authorization, set the CHECK IF HOST IS ONLINE ADDR VERIFY field to Y. Set this field to N if the BASE24-pos Authorization module does not perform address verification before sending transactions to a host for authorization. The BASE24-pos Authorization module performs address verification on transactions it authorizes, regardless of this field setting.

6. Update the CPF record by pressing the **F5** key.

CPF Screen 7

CPF screen 7 is shown below, followed by descriptions of the ADDRESS VERIFICATION ALGO and CHECK IF HOST IS ONLINE ADDR VERIFY fields.

```
BASE24-POS     CARD PREFIX            LLLL        YY/MM/DD  HH:MM  07 OF 22
PREFIX:                     PAN LENGTH: 00                  FIID:

                              POS INFORMATION

              MAXIMUM NUMBER OF REFUNDS:    0
       MAXIMUM NUMBER OF PRE-AUTH HOLDS:  2
                      CHARGEBACK UPDATE:  N
                   REPRESENTMENT UPDATE:  N
                DEFAULT COMBO CARD TYPE:  1
                   DEFAULT ACCOUNT TYPE: 00  (NO ACCOUNT)
                                         00  (NO ACCOUNT)
                                         00  (NO ACCOUNT)

                         POS PROCESSING INFORMATION
                                ISSUER:
                    PIN PROCESSING FLAG: 0 (PIN NOT REQUIRED)
              ADDRESS VERIFICATION ALGO: V  (VISANET)
 CHECK IF HOST IS ONLINE    ADDR VERIFY: N  (Y/N)

*********************************** BASE24 ***********************************
 NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                    F12-HELP
```

ADDRESS VERIFICATION ALGO — A code identifying the algorithm used for address verification. Valid values are as follows:

M = Banknet (MasterCard) algorithm
V = VisaNet algorithm

Field Length: 1 alphanumeric character
Required Field: Yes
Default Value: V

CHECK IF HOST IS ONLINE ADDR VERIFY — A code indicating whether address verification should be performed during prescreening for transactions containing address information. Valid values are as follows:

Y = Yes, perform address verification during prescreening.
N = No, do not perform address verification during prescreening.

Field Length: 1 alphanumeric character
Required Field: Yes
Default Value: N

Maintaining Cardholder Addresses on BASE24

Add-on BASE24-pos Address Verification module processing uses address and ZIP code data stored in the CAF. These addresses can be added, deleted, and changed using files maintenance procedures.

The following three options are available for maintaining cardholder address information in the CAF:

- CAF screen 12 displays address verification fields for online files maintenance.
- Address verification fields can be included in the refresh of a CAF record.
- Address verification fields can be included in updates using BASE24-from host maintenance processing.

Setting Up the Institution Definition File (IDF)

Institution Definition File (IDF) screen 5 contains flags that identify the segments to be included in files belonging to the institution. Before you can modify cardholder address information in the CAF, you must set the address verification flag on IDF screen 5.

To set the address verification flag in the IDF, proceed as follows:

1. Display IDF screen 1 by entering IDF in the FILE DESTINATION field at the bottom of the CRT access screen and pressing the **F1** key.
2. Enter the FIID of the financial institution in the FIID field, and press the F2 key.
3. Display IDF screen 5 by entering the value 5 in the NEW PAGE field at the bottom of the screen and pressing the **F7** key.
4. Enter the value Y next to the ADDRESS VERIFICATION file segment indicator.
5. Update the IDF by pressing the **F5** key.

IDF Screen 5

IDF screen 5 is shown below, followed by descriptions of its address verification fields.

```
BASE24-BASE  INSTITUTION FILE        LLLL       YY/MM/DD  HH:MM  05 OF 43

                 FIID:             FI-NAME:

                 FIID AUTH FILE SEGMENT INDICATORS

   Y  BASE                 N  ATM                  Y  POS
   N  TELLER               N  TELEBANKING          N  CREDIT LINE
   N  NAME (PBF SHORT)     N  SSB BASE             N  SSB CHECK
   Y  ADDRESS VERIFICATION N  CUST SRVC/FRAUD CNTL N  PRE-AUTH
   N  NON-CRNCY DISPENSE








********************************** BASE24 **********************************
 NEW PAGE:        FILE DESTINATION:       NEW LOGICAL NETWORK ID:
                         F12-HELP
```

FIID AUTH FILE SEGMENT INDICATORS — A series of flags indicating which segments this institution's authorization files contain.

These flags permit each institution to carry only the information needed for the BASE24 products it supports. Each segment requires additional disk space for each customer record in the authorization files. Disk space can be used more efficiently if each institution's authorization files contain only the segments used by that institution.

The ADDRESS VERIFICATION file segment is for institutions including BASE24-pos address verification information in CAF records.

Valid values for each indicator are as follows:

Y = Yes, include this segment.
N = No, do not include this segment.

The indicator for the base segment (BASE) is always set to a value of Y because that segment is included for all products. The POS segment is required (that is, set to a value of Y) before the Address Verification segment can be added.

Field Length:	1 alphabetic
Occurs:	2–32 times, depending on the number of segments supported by the logical network.
Required Field:	Yes
Default Value:	N (except for the Base segment indicator, which is always the value Y)
Data Name:	IDF.IDFBASE.FIID-SEG-MAP

For more information about the IDF and file segment indicators, refer to the ***BASE24 Base Files Maintenance Manual.***

Modifying Cardholder Address Information in the CAF

To add or change address verification information in the CAF online, proceed as follows:

1. Display CAF screen 1 by entering CAF in the FILE DESTINATION field at the bottom of the CRT access screen and pressing the **F1** key.
2. Read the CAF record being changed by completing the PAN, MEMBER, and FIID fields and pressing the **F2** key.
3. Display CAF screen 12 by entering 12 in the NEW PAGE field at the bottom of the CRT access screen and pressing the **F7** key.
4. Enter or change the information in the ADDRESS and ZIP CODE fields.
5. Update the CAF record by pressing the **F5** key.

CAF Screen 12

CAF screen 12 is shown below, followed by descriptions of its fields.

```
BASE24-POS   CARDHOLDER FILE          LLLL        YY/MM/DD  HH:MM  12 OF 21

        PAN:                                  MEMBER: 000   FIID:

                    ADDRESS VERIFICATION INFORMATION

    ADDRESS:
   ZIP CODE:            (MUST BE 5 OR 9 NUMERIC
                         CHARACTERS FOR U.S.
                         CARDHOLDERS)









********************************** BASE24 **********************************
 NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                        F12 - HELP
```

ADDRESS — The cardholder billing address used when performing address verification. All information that is numeric in nature must be entered as numeric data (e.g., First Street must be entered as 1st Street).

Field Length:	1–20 alphanumeric characters
Required Field:	No
Default Value:	No default value
Data Name:	CAF.AVCAF.ADDR

ZIP CODE — The cardholder billing ZIP code used when performing address verification. The ZIP codes of U.S. cardholders must be either 5 or 9 digits in length. The ZIP codes of non-U.S. cardholders can be one to nine alphanumeric characters in length. ZIP codes of less than nine characters must be left-justified. Unused positions must be blank.

Field Length:	1–9 alphanumeric characters
Required Field:	No
Default Value:	No default value
Data Name:	CAF.AVCAF.ZIP-CDE

Maintaining Cardholder Addresses at the Interchange

Cardholder addresses stored by the VisaNet or Banknet Interchange can also be updated using files maintenance procedures. The VisaNet File Update Request Log File (FRLF) enables access to address verification data stored and maintained by the VisaNet Interchange. The Banknet File Update Request Log File (BNRLF) enables access to address verification data stored and maintained by the Banknet Interchange.

Accessing Cardholder Address Information on VisaNet

Updates to cardholder address information are sent to the VisaNet Interchange using FRLF file update requests. To add or change address verification information stored and maintained by VisaNet, proceed as follows:

1. Display ICF screen 1 by entering ICF in the FILE DESTINATION field at the bottom of the CRT access screen and pressing the **F1** key.
2. Read the ICF record being changed by completing the INTERCHANGE FIID and PROCESS fields and pressing the **F2** key.
3. Display ICF screen 12 by entering 12 in the NEW PAGE field at the bottom of the CRT access screen and pressing the **F7** key.
4. Access the VisaNet function menu by pressing the **F8** key.
5. Select FRLF from the VisaNet function menu by placing the cursor next to the FRLF entry and pressing the **F1** key.
6. Read the FRLF record being changed by completing the SYMBOLIC NAME field or the RN NAME and PROCESS NUMBER fields and pressing the **F2** key.
7. Enter or change the information in the ADDRESS and ZIP CODE fields.
8. Send the FRLF update file request to VisaNet by pressing the **F5** key.

FRLF Screen 1

FRLF screen 1 is shown below, followed by descriptions of its address verification fields.

```
BASE24-BASE VISANET ACTIVITY MNT     LLLL        YY/MM/DD  HH:MM  01 OF 02


    PROCESS NUMBER:

    VISA SYSTEM: __         NETWORK ID: ____          FILE TYPE: _

    ACCT NUMBER: _______________________            COUNTRY CODE: _

    OPERATOR ID: ___       PURGE DATE: ______ (YYMMDD)  ACT CODE: __

    REGION CODE: _________                    SPENDING COUNT LIMIT: __

    SPENDING AMOUNT LIMIT: ______ CVA CODE: __         PVK INDEX: _

    PVV: ____     ADDRESS: ____________________  ZIP CODE: _________

 *********************************** BASE24 ********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                          F12-HELP
```

ADDRESS — The cardholder billing address used when performing address verification. All information that is numeric in nature must be entered as numeric data (e.g., First Street must be entered as 1st Street).

Field Length: 1–20 alphanumeric characters
Required Field: No
Default Value: No default value

ZIP CODE — The cardholder billing ZIP code used when performing address verification. The ZIP codes of U.S. cardholders must be either five or nine digits in length. The ZIP codes of non-U.S. cardholders can be from one to nine alphanumeric characters in length. ZIP codes of less than nine characters must be left-justified. Unused positions must be blank.

Field Length: 1–9 alphanumeric characters
Required Field: No
Default Value: No default value

Accessing Cardholder Address Information on Banknet

Cardholder address data at the Banknet Interchange can be added, updated, and deleted using BNRLF file update requests. The BNRLF can also display cardholder address data from the Banknet Interchange for operators to read. Each time a record from Banknet is added, updated, deleted, or read, the file update request is also logged to the BNRLF for later perusal.

Modifying Cardholder Address Information Using the BNRLF

To add, update, delete, or display address verification data stored and maintained by Banknet, proceed as follows:

1. Display ICF screen 1 by entering ICF in the FILE DESTINATION field at the bottom of the CRT access screen and pressing the **F1** key.
2. Read the ICF record being changed by completing the INTERCHANGE FIID and PROCESS fields and pressing the **F2** key.
3. Display ICF screen 12 by entering 12 in the NEW PAGE field at the bottom of the CRT access screen and pressing the **F7** key.
4. Access the Banknet menu by pressing the **F8** key.
5. Select BNRLF from the Banknet function menu by placing the cursor next to the BNRLF entry and pressing the **F1** key.
6. Access the address verification information by placing the cursor before the MCC105 AVS FILE field and pressing the **F1** key.
7. Add, update, delete, or display the address verification data by performing one of the following substeps:
   a. Add address verification data by completing the ACCOUNT NUMBER, SECURITY CODE, ADDRESS KEY, and ZIP CODE fields. Send the file update request to Banknet by pressing the **F3** key.
   b. Update address verification data by completing the ACCOUNT NUMBER, SECURITY CODE, ADDRESS KEY, and ZIP CODE fields, using the modified information where needed. Send the file update request to Banknet by pressing the **F5** key.
   c. Delete address verification data by completing the ACCOUNT NUMBER, SECURITY CODE, ADDRESS KEY, and ZIP CODE fields. Send the file update request to Banknet by pressing the **F4** key.

d. Read address verification data by completing the ACCOUNT NUMBER and SECURITY CODE fields and pressing the **F2** key.

Perusing Records in the BNRLF

To peruse file update requests submitted to Banknet and those records read using the BNRLF, proceed as follows:

1. Display ICF screen 1 by entering ICF in the FILE DESTINATION field at the bottom of the CRT access screen and pressing the **F1** key.
2. Read the ICF record being changed by completing the INTERCHANGE FIID and PROCESS fields and pressing the **F2** key.
3. Display ICF screen 12 by entering 12 in the NEW PAGE field at the bottom of the CRT access screen and pressing the **F7** key.
4. Access the Banknet menu by pressing the **F8** key.
5. Select BNRLF from the Banknet function menu by placing the cursor next to the BNRLF entry and pressing the **F1** key.
6. Access the BNRLF perusal screen by entering 3 in the NEW PAGE field at the bottom of the CRT access screen and pressing the **F7** key.
7. Begin reading BNRLF records by completing either the PERUSAL DATE (YYMMDD) or the PAN field and pressing the **F1** key. When the **F1** key is pressed without completing either the PERUSAL DATE (YYMMDD) or the PAN field, the first record in the BNRLF is shown on the screen.

   Note: Records are logged chronologically and, therefore, must be perused chronologically. For example, the first record read might be from the MCC109 RCL/CWB file and the second record read might be from the MCC105 AVS file.
8. Continue reading BNRLF records as needed by pressing the **F6** key to display subsequent records.

BNRLF Screen 1

BNRLF screen 1 is shown below. This screen is a menu screen from which three different subfiles can be accessed. The MCC105 AVS File is used to access address verification information stored and maintained by the Banknet Interchange.

```
 BASE24-BNET  FILE REQUST SELECTION   LLLL       YY/MM/DD  HH:MM  01 OF 03



                   _   MCC102 ACCOUNTS FILE

                   _   MCC105 AVS FILE

                   _   MCC103 ACCOUNTS MANAGEMENT FILE





********************************** BASE24 *********************************
 NEW PAGE:       FILE DESTINATION:        NEW LOGICAL NETWORK ID:
F1-ENTER   F10-PRINT   F12-HELP  F16-EXIT
```

MCC105 AVS File Screen (BNRLF Screen 2)

The MCC105 AVS File screen (BNRLF screen 2) is shown below, followed by descriptions of its fields.

```
BASE24-BNET  MCC105 FILE REQUEST      LLLL       YY/MM/DD  HH:MM  02 OF 03



    ACCOUNT NUMBER: ___________________    SECURITY CODE: _____

                       ADDRESS DATA

                  ADDRESS KEY: _____

                     ZIP CODE: _________

                SERVER SUFFIX: _


********************************** BASE24 *********************************
 NEW PAGE:       FILE DESTINATION:       NEW LOGICAL NETWORK ID:
                       F12-HELP
```

ACCOUNT NUMBER — The primary account number of the cardholder whose address information is being requested. Only numeric characters are valid.

Field Length:	1–19 numeric characters
Required Field:	Yes
Default Value:	No default value

SECURITY CODE — A Banknet code that must be entered when requesting a BNRLF file update. This code indicates that the user has proper security access to update or request cardholder address data maintained by Banknet.

Field Length:	5 numeric characters
Required Field:	Yes
Default Value:	No default value

ADDRESS KEY — The 5-digit numeric combination resulting from the Banknet address verification algorithm.

Field Length:	5 numeric characters
Required Field:	Yes
Default Value:	No default value

ZIP CODE — The cardholder billing ZIP code used when performing address verification. The ZIP codes of U.S. cardholders must be either five or nine digits in length. The ZIP codes of non-U.S. cardholders can be one to nine alphanumeric characters in length. ZIP codes of less than nine characters must be left-justified. Unused positions must be blank.

Field Length:	1–9 alphanumeric characters
Required Field:	Yes
Default Value:	No default value

SERVER SUFFIX — The server suffix used for sending the message. The server suffix determines which interface sends the transaction. Suffixes used are specific to each system's configuration. Valid values are 0 through 9 or blank.

Field Length:	1 numeric character
Required Field:	No
Default Value:	No default value

BNRLF File Request Perusal Screen (BNRLF Screen 3)

The BNRLF File Request Perusal screen (BNRLF screen 3) is shown below, followed by descriptions of its fields. This screen applies to MCC102, MCC105, and MCC109 perusals.

```
 BASE24-BNET  FILE REQUEST PERUSAL     LLLL       YY/MM/DD  HH:MM  03 OF 03

                       MCC105 FILE REQUEST

 PERUSAL DATE(YYMMDD): ______         PAN: ____________________________

                          TIME: ________
                 RESPONSE CODE: __
                 SECURITY CODE: _____
                    PIN LENGTH: __
                   ADDRESS KEY: _____
                      ZIP CODE: _________





********************************** BASE24 *********************************
 NEW PAGE:       FILE DESTINATION:       NEW LOGICAL NETWORK ID:
F1-READ    F4-DELETE   F6-READ NEXT    F10-PRINT    F16-EXIT
```

PERUSAL DATE — The date of the file update request and response being perused. If this field and the PAN field are blank, the first record in the file is displayed.

Field Length:	6 numeric characters in YYMMDD format
Required Field:	No
Default Value:	The date of the first record in the file.

PAN — The PAN of the updated file being perused. If this field and the PERUSAL DATE field are blank, the first record in the file is displayed.

Field Length:	1–28 numeric characters
Required Field:	No
Default Value:	The PAN of the first record in the file.

TIME — The time the update request and response was logged to the BNRLF.

Field Length: System-protected

RESPONSE CODE — A code from Banknet indicating the response to the MCC105 file update request. Valid values are as follows:

00 = File update action completed successfully
25 = Unable to locate record on file (no action taken)
27 = File update field edit error
30 = Format error
40 = Requested function not supported
63 = Security violation
80 = Duplicate add; action not performed
96 = System error

Field Length: System-protected

SECURITY CODE — The Banknet security code associated with the file request. The security code was originally entered with the file request to ensure that the user had proper security access to update the file.

Field Length: System-protected

PIN LENGTH — The PIN length associated with the file request.

Field Length: System-protected

ADDRESS KEY — The five-digit numeric combination resulting from the Banknet address verification algorithm.

Field Length: System-protected

ZIP CODE — The cardholder billing ZIP code used when performing address verification. The ZIP codes of U.S. cardholders must be either five or nine digits in length. The ZIP codes of non-U.S. cardholders can be one to nine alphanumeric characters in length. ZIP codes of less than nine characters must be left-justified. Unused positions must be blank.

Field Length: System-protected

Perusing Cardholder Transactions

BASE24-pos reports and log files include address verification information for inquiry and reference purposes. The Daily Cardholder Activity Report (B24POS02) and Daily Retailer Activity Report (B24POS12) contain three additional fields for each transaction involving address verification—address, ZIP code, and status code. The address and ZIP code on the reports are the ones received with the transaction. The status code on the reports is the final result of address verification processing.

The POS Transaction Log File (PTLF) includes the address verification address, ZIP code, and status code fields. These fields are included in the transactions available for online perusal using the PTLF perusal function.

Refer to the ***BASE24-pos Files Maintenance Manual*** for information about perusing the PTLF.

3: Host Impacts

The host impacts resulting from BASE24-pos Address Verification processing vary from site to site, depending on the information exchanged between BASE24-pos and a particular host during transaction processing, reporting, and managing the BASE24-pos database. BASE24-pos and a host use the BASE24-pos Host Interface for exchanging transaction information. The BASE24 Super Extract process provides BASE24-pos reporting information to a host. BASE24-from host maintenance and Refresh provide host database information to BASE24-pos.

This section identifies the impacts BASE24-pos Address Verification has on host processing and explains how to obtain the refresh record layout for the Address Verification segment of the Cardholder Authorization File (CAF).

Host impacts involving BASE24-from host maintenance and Refresh occur only when the BASE24-pos add-on Address Verification module is used. All other host impacts presented in this section occur with the standard BASE24-pos product or both the standard product and the add-on module.

Host Interface Process

The external messages used by ANSI and ISO Host Interface processes can be configured for each host to include or exclude address verification data. BASE24-pos Host Interface configuration procedures are provided in section 2.

- ANSI External Message. The BASE24-pos Standard External Message (PSEM) is supported in regular and extended formats, with the extended format containing ADDR, ZIP-CDE, and ADDR-VRFY-STAT fields in addition to the regular format fields. The PSEM is documented in the ***BASE24-pos Transaction Processing Manual***.
- ISO External Message. Two additional elements have been defined in the ISO external message for address verification data. Element P-44 contains the address verification status and element P-63 contains the address and ZIP code information in the Address Verification Token. Element P-44 requires special host processing to set the value of its Responder Identity field if address verification is performed on the host. Element P-63 requires special host processing to handle its variable token contents.

Super Extract Process

Address verification data has been included in the extracts of four files, as described below. If these files are used in host processing, the host applications must be changed to handle the new data. The DDLs contain additional details.

- Interchange Log File (ILF), Position 612 in the POS message contains the ADDR-VRFY-STAT field, which carries the status code. The status code indicates the result of performing address verification.
- POS Transaction Log File (PTLF). The PTLF Financial Transaction record contains the ADDR, ZIP-CDE, and ADDR-VRFY-STAT fields. These fields are in binary/ASCII and ASCII-only PTLF record formats.
- Online Maintenance File (OMF). The Address Verification segment is included in an OMF record written with CAF record information, increasing the record length by 60 bytes. The ADDR-VRFY and ADDR-VRFY-ALGO fields are included in an OMF record written with CPF record information, but the record length is not changed.
- Update Log File (ULF). The Address Verification segment is included in a ULF record written with CAF record information, increasing the record length by 60 bytes.

BASE24-from host maintenance Product

The CAF contains address and ZIP code information used by the add-on BASE24-pos Address Verification module to verify addresses. The CAF information can be obtained from the host using ANSI or ISO messages using BASE24-from host maintenance. The host configuration procedures in section 2 include BASE24-from host maintenance.

- ANSI External Message. The BASE24-from host maintenance ANSI external message contains the address and ZIP code information in the CAF Update Request message. The ANSI external message is documented in the ***BASE24 External Message Manual***.
- ISO External Message. Element S-127 contains the address and ZIP code information. The ISO external message is documented in the ***BASE24 External Message Manual***.

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